

Fossil diatoms in Arctic marine surface sediments

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Diatoms are one of the main constituents of marine phytoplankton in the Arctic, and thanks to their siliceous skeletons, diatom fossils are relatively well preserved in sediments. Due to their species-specific sensitivity to different ocean conditions, their abundance and assemblages in sediments are routinely used by paleo-oceanographers to reconstruct the state of the surface ocean in the past (Koç 2007). By using statistical methods, we can obtain valuable knowledge about their ecological preferences (Oksman et al. 2019) and generate quantitative reconstructions of various parameters, such as sea-surface temperature and sea-ice concentration through time (Krawczyk et al. 2021; Sha et al. 2014). The Marine Arctic Diatoms (MARDI) working group (WG) (pastglobal-changes.org/mardi) aims to advance knowledge on marine-diatom ecology and diatom-based reconstructions by compiling and harmonizing data from surface-sediment samples across the Arctic.

The MARDI WG has collected diatom-assemblage data of more than 1000 marine-sediment surface samples from high latitude locations of the northern hemisphere (Fig. 1). Most of the data has been published, but it also includes a few newer, unpublished datasets. The data is sourced from different research groups, which means there may be differences in applied methodologies and diatom taxonomy, but we hypothesize that there are common trends in the datasets which allow us to identify relationships between assemblages and modern environmental data.

Following a successful workshop on Arctic marine diatom taxonomy in 2023, the group organized a 2024 workshop in Aarhus that focused on data compilation and analysis. The workshop brought together experts in diatom taxonomy and Arctic paleoceanography with data scientists specializing in statistical methods in paleoecology. A total of 14 participants from eight countries, ranging from graduate students to senior academic staff attended the workshop at the Department of Geoscience of Aarhus University. After a short round of introductions, participants spent three days discussing, reformatting and aligning the different datasets into a machine-readable format that can be integrated into a single

training set. Each diatom species was assigned a unique MARDI species code and compiled into a table, together with possible synonyms and correct species names, following the latest updated taxonomy. Each individual sample was assigned a MARDI site code and compiled into a separate table together with available metadata, which was mined from the original publications in many cases. This includes the original sample codes, latitude, longitude, water depth, collection date, sampling device, chronological information, publications, and any other available information for each sample. The spatial range of the surface samples includes large parts of the North Pacific and North Atlantic Ocean, including the coastal waters around Greenland (Fig. 1). In the Central Arctic Ocean, the samples occur in two clusters in the Laptev Sea and Chukchi Sea. At the end of the workshop, diatom counts from all surface-sediment samples, sourced from various research groups and publications, were aligned into a common format.

In addition to the diatom counts, another discussion focused on which environmental parameters play a role in diatom ecology and would, therefore, be relevant to include in the datasets. Workshop participants queried various databases

containing modern environmental observations from satellites and in situ measurements, including sea-surface temperature, sea-ice cover, nutrients, and salinity, to retrieve environmental data for each site in the MARDI training set.

The culmination of the three-day event resulted in the creation of R scripts (v. 3.3; R core team 2024) capable of importing and harmonizing the data from each different dataset contribution, representing a major deliverable of the group. Over the coming months, the workshop participants will continue their collaboration through short online meetings focusing on exploring the statistical relationships between the surface-sediment training set and environmental parameters.

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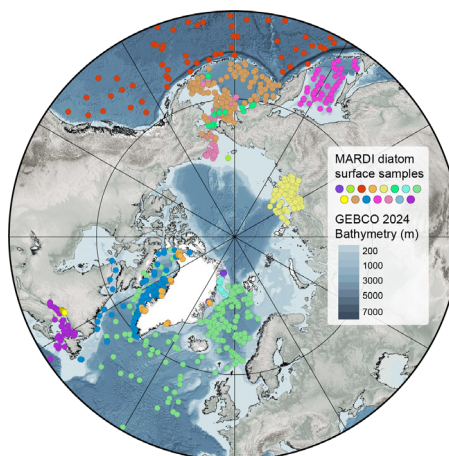


Figure 1: Map of the high latitudes of the Northern Hemisphere showing the spatial distribution of surface samples in the MARDI diatom dataset. Each dot represents a surface sample, and the different colors indicate the 15 different datasets used as input during the workshop. Background map from GEBCO Bathymetric Compilation Group (2024).